Middle East Technical University
Department of Electrical and Electronics Engineering

EE201 – Circuit Theory I

Lecture 9: Trees, Cutsets and Fundamental Loops

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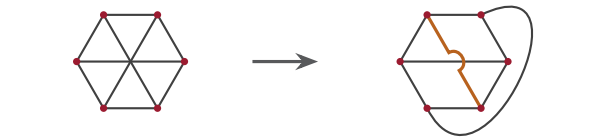
Colour code used in these notes and in the slides

current i | voltage v, cutsets, loops | tree branches | nodes ①

1. Beyond planar graphs

Why this lecture? Node voltages work for any connected graph, with one fixed reference node. Mesh currents need a planar, nonseparable graph. The goal of this lecture is methods that work for every connected graph, including nonplanar and separable ones. The key is a **tree**. Its fundamental cutsets generalise the node equations (the same $n-1$ count, but the tree can be chosen). Its fundamental loops generalise the mesh equations and work for every connected graph.

Meshes need a drawing with no crossings. A nonplanar graph has no such drawing, so it has no meshes and no mesh matrix. In the hexagon below, the three diagonals cross at the centre. Moving one diagonal outside removes one crossing, but a jumper wire (a hop) is still needed.



crossings at the centre jumper wire: still one hop

2. Trees

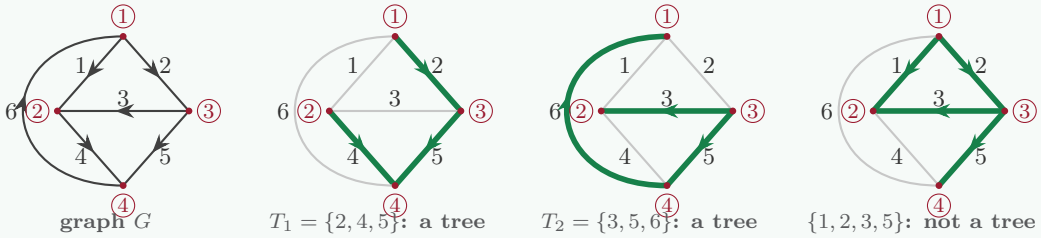
Tree

Let G be a connected graph. A subgraph T of G is a **tree** if

1. T contains all the nodes of G,
2. T is connected, and
3. T contains no loops.

Example 9.1 — trees of the bridge graph

Which of the green subgraphs are trees?



T_1 and T_2 reach all four nodes, are connected and have no loop. The set $\{1,2,3,5\}$ contains the loop of branches 1, 2, 3, so it is not a tree. A graph has many trees.

A tree has $n-1$ branches

Build the tree one branch at a time, starting from a single node. Each new branch must reach a new node: a branch between two nodes already reached would close a loop. All n nodes must be reached, so the tree has exactly $n-1$ branches. The other $b-n+1$ branches are left out.

3. Cutsets

Cutset

Let G be a connected graph. A set C of branches is a **cutset** if

1. removing all the branches of C leaves the graph unconnected, and
2. after removing C, putting back a single branch of C makes the graph connected again, *whichever* branch of C is put back.

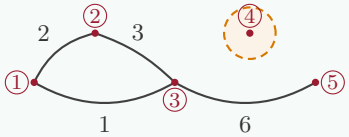
Removing a cutset leaves exactly two parts. A closed surface around one part crosses exactly the branches of C.

Condition 2 makes the cutset minimal: every branch is needed. Minimal does not mean fewest; cutsets of one graph can differ in size.

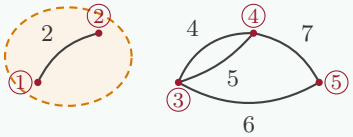
Conversely, a closed surface gives a cutset only if the nodes inside, and those outside, are each connected among themselves.

Example 9.2 — cutsets of a figure-eight graph

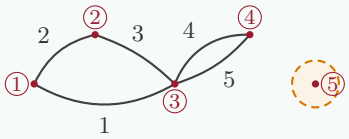
In each panel the removed branches are deleted; look at what is left.



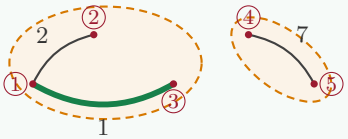
$C_1 = \{4, 5, 7\}$: node ④ cut off



$C_2 = \{1, 3\}$: nodes ①, ② cut off



$C_3 = \{6, 7\}$: node ⑤ cut off



$\{1, 3, 4, 5, 6\}$ with 1 put back: still two pieces

For C_1: removing branches 4, 5, 7 cuts off node ④, so condition 1 holds. Putting back each one of them reaches node ④ again, so condition 2 holds. C_2 and C_3 are checked the same way.

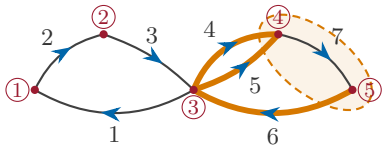
The set $\{1, 3, 4, 5, 6\}$ is *not* a cutset: removing it leaves three pieces, and with branch 1 put back the graph is still unconnected (condition 2 fails). $\{4, 5, 6\}$ alone is.

Condition 2 must hold *whichever* branch is put back. Removing $\{1, 4, 5, 7\}$ cuts off node ④; putting back 4, 5 or 7 reconnects it, but putting back 1 does not. So it is not a cutset; $C_1 = \{4, 5, 7\}$ is.

The set $\{4, 6\}$ fails the other way: removing it leaves the graph connected, since branch 5 still joins ③ and ④ (condition 1 fails). Branches 4 and 5 are parallel, so a closed surface crosses both or neither: parallel branches are in a cutset together or not at all.

KCL on a cutset. By KCL on the closed surface around one part, the currents of the cutset branches add up to zero (each with + if it leaves the surface, − if it enters).

Take the branch currents below, in the arrow directions. They satisfy KCL at every node. Consider the cutset $\{4, 5, 6\}$, with the surface around nodes ④ and ⑤.



branch	1	2	3	4	5	6	7
current (A)	2	2	2	1	2	3	3

Current i_6 leaves the surface; i_4 and i_5 enter it:

$$i_6 - i_4 - i_5 = 3 - 1 - 2 = 0.$$

This is the sum of KCL at nodes ④ and ⑤: $(i_7 - i_4 - i_5) + (i_6 - i_7) = 0$. The current i_7 is inside the surface and cancels.

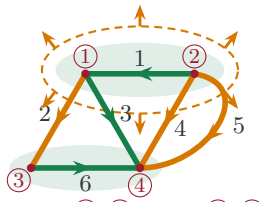
4. Fundamental cutsets

Fix a tree T of a connected graph G. The branches of G that are not in T form the **co-tree**. Now pick one tree branch β and remove it. The tree falls into two parts. The branches of G that join these two parts form a cutset.

Fundamental cutset

The **fundamental cutset** of a tree branch β is

- β plus the co-tree branches that join the two parts of the tree;
- it contains only one tree branch, β;
- its direction is that of β.



parts {①, ②} and {③, ④}

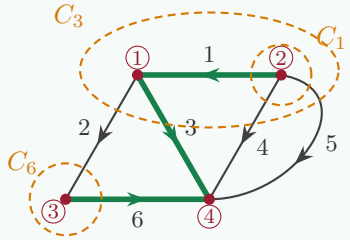
For the graph G of Example 9.3 below, take $\beta = 3$. Removing it leaves the parts {①, ②} and {③, ④}, and the fundamental cutset is $\{3, 2, 4, 5\}$.

One tree branch. Every other tree branch (1, 6) lies inside one part, so it does not join them.

Direction. Branch 3 leaves the surface around {①, ②}, so the cutset points out of it (arrows). In KCL a branch counts + if it crosses the surface the same way as β, and − otherwise.

Example 9.3 — fundamental cutsets and KCL

The graph G has $n = 4$ nodes and $b = 6$ branches. Take the tree $\{1, 3, 6\}$ (green, $n - 1 = 3$ branches). The co-tree is $\{2, 4, 5\}$ ($b - n + 1 = 3$ branches).



tree br.	cutset	KCL
1	$\{1, 4, 5\}$	$i_1 + i_4 + i_5 = 0$
3	$\{3, 2, 4, 5\}$	$i_3 + i_2 + i_4 + i_5 = 0$
6	$\{6, 2\}$	$i_6 - i_2 = 0$

Each surface encloses one of the two parts left after removing a tree branch. For C_3: removing branch 3 leaves the parts {①, ②} and {③, ④}. Branches 3, 2, 4, 5 join them, and all four go from the upper part to the lower part, the same way as branch 3: every term has +.

KCL on the fundamental cutsets

A tree gives $n - 1$ fundamental cutsets, so $n - 1$ KCL equations. They are independent: each equation contains its own tree branch current, which appears in no other equation.

Branch currents from co-tree currents. We can relate every branch current to the co-tree currents. A co-tree current is itself. Each fundamental cutset has one tree branch, so its KCL gives that tree branch current in terms of co-tree currents. For G:

$$i_1 = -i_4 - i_5, \quad i_3 = -i_2 - i_4 - i_5, \quad i_6 = i_2.$$

So all b branch currents are combinations of the $b - n + 1$ co-tree currents, and KCL holds automatically, like mesh currents.

Matrix form. The fundamental cutset equations can be collected in the **fundamental cutset matrix** Q: one row per tree branch, one column per branch. An entry is $+1$ or -1 if the branch is in the cutset, in the same or the opposite direction, and 0 otherwise. For G:

$$Q = \begin{array}{c} \\ 1 \\ 3 \\ 6 \end{array} \begin{array}{c} \begin{array}{cccccc} 1 & 2 & 3 & 4 & 5 & 6 \end{array} \\ \left[\begin{array}{cccccc} \boxed{1} & 0 & \boxed{0} & 1 & 1 & \boxed{0} \\ \boxed{0} & 1 & \boxed{1} & 1 & 1 & \boxed{0} \\ \boxed{0} & -1 & \boxed{0} & 0 & 0 & \boxed{1} \end{array}\right] \end{array}, \qquad Q\mathbf{i} = \mathbf{0} \quad (\text{like } A\mathbf{i} = \mathbf{0}).$$

The tree columns (boxed) form an identity, so the rows are independent. The relation above, in matrix form, is

$$\begin{bmatrix} i_1 \\ i_2 \\ i_3 \\ i_4 \\ i_5 \\ i_6 \end{bmatrix} = \underbrace{\begin{bmatrix} 0 & -1 & -1 \\ 1 & 0 & 0 \\ -1 & -1 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}}_{B^T} \begin{bmatrix} i_2 \\ i_4 \\ i_5 \end{bmatrix}, \qquad \mathbf{i} = B^T \mathbf{i}_C \quad (\text{like } \mathbf{i} = M^T \mathbf{i}_M).$$

Its transpose B, with rows for the co-tree branches 2, 4, 5, comes back with the fundamental loops.

5. Fundamental loops

Fix a tree T of a connected graph G and pick one co-tree branch. Adding it to the tree closes exactly one loop: the co-tree branch plus the tree path between its two ends. For example, adding co-tree branch 4 to the tree $\{1, 3, 6\}$ of G closes the loop $\{4, 3, 1\}$.

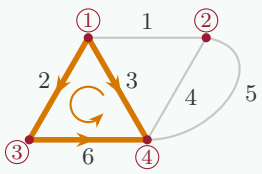
Fundamental loop

The **fundamental loop** of a co-tree branch is

- that co-tree branch plus the tree path between its ends;
- it contains only one co-tree branch;
- its direction is that of the co-tree branch.

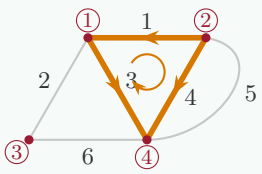
Example 9.4 — fundamental loops and KVL

Same graph and tree as in Example 9.3. Each co-tree branch closes one loop. Walk it in the direction of its co-tree branch: a branch voltage counts + if its arrow points along the walk, − if against.



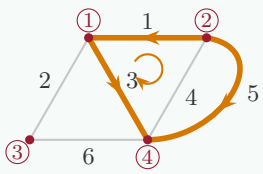
loop of 2: $\{2, 6, 3\}$

$v_2 + v_6 - v_3 = 0$



loop of 4: $\{4, 3, 1\}$

$v_4 - v_3 - v_1 = 0$



loop of 5: $\{5, 3, 1\}$

$v_5 - v_3 - v_1 = 0$

KVL on the fundamental loops

A tree gives $b-n+1$ fundamental loops, so $b-n+1$ KVL equations. They are independent: each equation contains its own co-tree branch voltage, which appears in no other equation.

Branch voltages from tree voltages. We can relate every branch voltage to the tree voltages. A tree voltage is itself. Each fundamental loop has one co-tree branch, so its KVL gives that co-tree branch voltage in terms of tree voltages. For G:

$$v_2 = v_3 - v_6, \quad v_4 = v_1 + v_3, \quad v_5 = v_1 + v_3.$$

So all b branch voltages are combinations of the $n-1$ tree voltages, and KVL holds automatically, like node voltages.

Matrix form. The fundamental loop equations are collected in the **fundamental loop matrix** B: one row per co-tree branch; an entry is $+1$ or -1 if the branch is in the loop, along or against its direction. It is the same B as in $\mathbf{i} = B^T \mathbf{i}_C$:

$$B = \begin{array}{c} \\ 2 \\ 4 \\ 5 \end{array} \begin{array}{c} \begin{array}{cccccc} 1 & 2 & 3 & 4 & 5 & 6 \end{array} \\ \begin{bmatrix} 0 & 1 & -1 & 0 & 0 & 1 \\ -1 & 0 & -1 & 1 & 0 & 0 \\ -1 & 0 & -1 & 0 & 1 & 0 \end{bmatrix} \end{array}, \qquad B\mathbf{v} = \mathbf{0} \quad (\text{like } M\mathbf{v} = \mathbf{0}).$$

The relation above, in matrix form, uses the transpose of the fundamental cutset matrix:

$$\begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \\ v_5 \\ v_6 \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -1 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}}_{Q^T} \begin{bmatrix} v_1 \\ v_3 \\ v_6 \end{bmatrix}, \qquad \mathbf{v} = Q^T \mathbf{v}_T \quad (\text{like } \mathbf{v} = A^T \mathbf{e}).$$

Finally $QB^T = \mathbf{0}$, like $AM^T = \mathbf{0}$.

6. Matrix form side by side

	incidence / mesh	fundamental cutset / loop
KCL	$A\mathbf{i} = \mathbf{0}$	$Q\mathbf{i} = \mathbf{0}$
KVL	$M\mathbf{v} = \mathbf{0}$	$B\mathbf{v} = \mathbf{0}$
$n-1$ voltages	$\mathbf{v} = A^T\mathbf{e}$ (node voltages)	$\mathbf{v} = Q^T\mathbf{v}_T$ (tree voltages)
$b-n+1$ currents	$\mathbf{i} = M^T\mathbf{i}_M$ (mesh currents)	$\mathbf{i} = B^T\mathbf{i}_C$ (co-tree currents)
orthogonality	$AM^T = \mathbf{0}$	$QB^T = \mathbf{0}$
KCL currents	$\mathcal{N}(A) = \mathcal{R}(M^T)$	$\mathcal{N}(Q) = \mathcal{R}(B^T)$
KVL voltages	$\mathcal{N}(M) = \mathcal{R}(A^T)$	$\mathcal{N}(B) = \mathcal{R}(Q^T)$
works for	A: any; M: planar, nonseparable	any connected graph

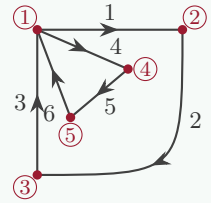
A circuit has $2b$ branch unknowns. A tree reduces them: the $n-1$ **tree branch voltages** fix every branch voltage (like node voltages), and the $b-n+1$ **co-tree branch currents** fix every branch current (like mesh currents), for any connected graph. The two columns describe the same two subspaces: $\mathcal{N}(A) = \mathcal{N}(Q)$ holds all currents that satisfy KCL, and $\mathcal{R}(A^T) = \mathcal{R}(Q^T)$ all voltages that satisfy KVL (Appendix B).

7. Any connected graph

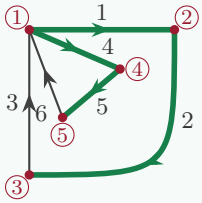
The tree is the key, and every connected graph has one. Its $n-1$ tree branches give the KCL equations, and its $b-n+1$ co-tree branches give the KVL equations. No drawing is needed. This works for nonplanar graphs, and it is also the fix for separable graphs met earlier.

Example 9.5 — the separable graph

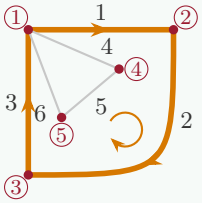
Two loops joined only at node ①. Previously the mesh method was not safe here. We follow the same steps as for G.



graph: $n=5$, $b=6$

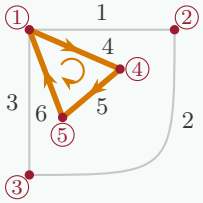


tree $\{1,2,4,5\}$



loop of 3: $\{3,1,2\}$

$v_3+v_1+v_2=0$



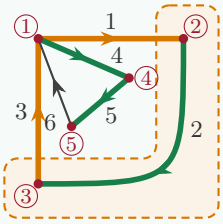
loop of 6: $\{6,4,5\}$

$v_6+v_4+v_5=0$

Step 1 — tree. Take the tree $\{1,2,4,5\}$ (green, $n-1=4$ branches). The co-tree is $\{3,6\}$ ($b-n+1=2$ branches).

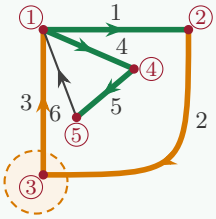
Step 2 — fundamental loops (KVL). Adding co-tree branch 3 closes the outer loop; adding co-tree branch 6 closes the inner triangle. Each is a single loop, walked in the direction of its co-tree branch (right-hand panels).

Step 3 — fundamental cutsets (KCL). Removing a tree branch splits the tree into two parts; the dashed surface encloses one of them, and the branches that cross it (orange) form the fundamental cutset. For tree branch 1 the parts are $\{①,④,⑤\}$ and $\{②,③\}$. Branches 1 and 3 cross the surface around $\{②,③\}$: 1 enters it, 3 leaves it, so $i_1-i_3=0$. Tree branches 2 and 5 cut off a single node (③, ⑤), and tree branch 4 cuts off the pair $\{④,⑤\}$.



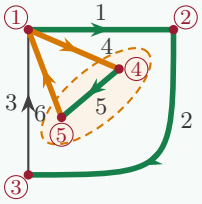
cutset of 1: $\{1,3\}$

$i_1-i_3=0$



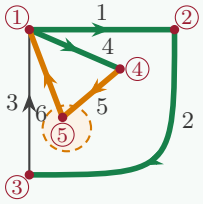
cutset of 2: $\{2,3\}$

$i_2-i_3=0$



cutset of 4: $\{4,6\}$

$i_4-i_6=0$



cutset of 5: $\{5,6\}$

$i_5-i_6=0$

Step 4 — result. The KCL equations give $i_1=i_2=i_3$ and $i_4=i_5=i_6$: one current for each loop, the co-tree currents i_3 and i_6. In total $4+2=6=b$ independent equations. Node ① causes no trouble.

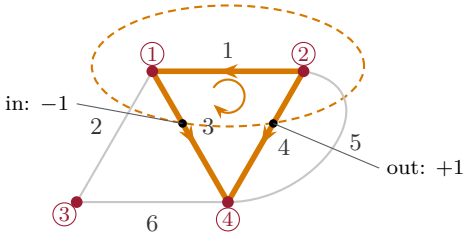
Summary

- A **tree** of a connected graph contains all nodes, is connected and has no loop. It has $n-1$ branches; the other $b-n+1$ form the **co-tree**.
- A **cutset** is a minimal set of branches whose removal cuts the graph in two. KCL holds on every cutset: it is KCL on a closed surface.
- Each tree branch has a **fundamental cutset**: it and some co-tree branches. These give $n-1$ independent KCL equations. Every branch current follows from the co-tree currents.
- Each co-tree branch has a **fundamental loop**: it and a tree path. These give $b-n+1$ independent KVL equations. Every branch voltage follows from the tree voltages.

Appendix A: why the rows of Q and B are orthogonal

From the matrix forms. $\mathbf{v} = Q^T \mathbf{v}_T$ satisfies KVL for every choice of tree voltages, so $BQ^T \mathbf{v}_T = \mathbf{0}$ for every $\mathbf{v}_T$. Hence $BQ^T = \mathbf{0}$, i.e. $QB^T = \mathbf{0}$. This is how $AM^T = \mathbf{0}$ was found previously.

By counting. The argument for a node and a mesh works with a closed surface in place of the node. Row $\mathbf{q}_k$ is the cutset of tree branch k: $+1$ for a branch that crosses its surface in the cutset direction, -1 the other way. Row $\mathbf{b}_r$ is a loop. Only loop branches that cross the surface give nonzero terms.



cutset of 3 and loop of 4 in G

Walk around the loop. A crossing in the cutset direction gives $+1$, a crossing the other way -1, whatever the branch arrow (an arrow against the walk flips both signs). The walk ends where it started, so it crosses as often one way as the other: $\mathbf{q}_k^T \mathbf{b}_r = 0$.

In G, the loop of 4 leaves the surface of the cutset of 3 on branch 4 and comes back on branch 3 (the cutset points out of the surface): $(+1)(+1) + (+1)(-1) = 0$. A node is the simplest closed surface, so the same counting also gives $AM^T = \mathbf{0}$.

Appendix B: null and range spaces of A, Q, M, B

Two subspaces of a matrix. Let X have b columns.

- The **null space** $\mathcal{N}(X) = \{\mathbf{x} \in \mathbb{R}^b : X\mathbf{x} = \mathbf{0}\}$: the vectors that X sends to zero.
- The **range space** $\mathcal{R}(X^T) = \{X^T\mathbf{y}\}$: all combinations of the rows of X. Its dimension is the rank of X.
- **Rank–nullity:** $\dim \mathcal{N}(X) = b - \operatorname{rank} X$.

The ranks. All four matrices have independent rows:

matrix	size	rank	why the rows are independent
A	$(n-1) \times b$	$n-1$	connected graph (shown earlier)
Q	$(n-1) \times b$	$n-1$	identity in the tree columns
M	$\ell \times b$	$\ell = b-n+1$	one row per mesh (planar, nonseparable)
B	$\ell \times b$	$b-n+1$	identity in the co-tree columns

So $\dim \mathcal{N}(A) = \dim \mathcal{N}(Q) = b - (n-1)$ and $\dim \mathcal{N}(M) = \dim \mathcal{N}(B) = b - (b-n+1) = n-1$.

Step 1 — currents: $\mathcal{N}(A) = \mathcal{R}(M^T)$.

- KCL says $\mathbf{i} \in \mathcal{N}(A)$. Mesh currents give $\mathbf{i} = M^T\mathbf{i}_M \in \mathcal{R}(M^T)$.
- $AM^T = \mathbf{0}$, so $A(M^T\mathbf{i}_M) = \mathbf{0}$: every vector of $\mathcal{R}(M^T)$ lies in $\mathcal{N}(A)$.
- Both have dimension $b-n+1$. A subspace inside another of the same dimension is all of it, so $\mathcal{R}(M^T) = \mathcal{N}(A)$.

Meaning: *every* current that satisfies KCL is $M^T\mathbf{i}_M$ for exactly one choice of mesh currents (the rows of M are independent). Mesh currents miss nothing and repeat nothing. The same argument with $QB^T = \mathbf{0}$ gives $\mathcal{N}(Q) = \mathcal{R}(B^T)$: co-tree currents.

Step 2 — voltages: $\mathcal{N}(M) = \mathcal{R}(A^T)$. KVL says $\mathbf{v} \in \mathcal{N}(M)$, and node voltages give $\mathbf{v} = A^T\mathbf{e}$. Now $MA^T = (AM^T)^T = \mathbf{0}$, and both subspaces have dimension $n-1$. So $\mathcal{N}(M) = \mathcal{R}(A^T)$: every voltage that satisfies KVL comes from exactly one set of node voltages. Likewise $\mathcal{N}(B) = \mathcal{R}(Q^T)$: tree voltages.

Step 3 — A and Q give the same KCL. Each row of Q is KCL on a closed surface, $\pm$ the sum of the node KCL equations inside it (take the part without the reference node). So every row of Q is a combination of rows of A: $Q = TA$ for some $(n-1) \times (n-1)$ matrix T. Since Q has rank $n-1$, T is invertible. Hence $Q\mathbf{i} = \mathbf{0} \iff A\mathbf{i} = \mathbf{0}$, i.e. $\mathcal{N}(Q) = \mathcal{N}(A)$, and also $\mathcal{R}(Q^T) = \mathcal{R}(A^T)$. With Steps 1 and 2, all the equalities in the matrix-form table follow; for example $\mathcal{N}(M) = \mathcal{R}(A^T) = \mathcal{R}(Q^T) = \mathcal{N}(B)$.

The picture. $\mathbb{R}^b$ splits into two orthogonal subspaces:

	KCL currents	KVL voltages
as a null space	$\mathcal{N}(A) = \mathcal{N}(Q)$	$\mathcal{N}(M) = \mathcal{N}(B)$
as a range space	$\mathcal{R}(M^T) = \mathcal{R}(B^T)$	$\mathcal{R}(A^T) = \mathcal{R}(Q^T)$
unknowns	mesh or co-tree currents	node or tree voltages
dimension	$b-n+1$	$n-1$

The dimensions add up to b. The subspaces are orthogonal: take any $\mathbf{i}$ that satisfies KCL and any $\mathbf{v}$ that satisfies KVL, so $\mathbf{v} = A^T\mathbf{e}$ by Step 2. Then

$$\mathbf{v}^T\mathbf{i} = (A^T\mathbf{e})^T\mathbf{i} = \mathbf{e}^T(A\mathbf{i}) = 0.$$

This is **Tellegen's theorem**: $\sum_k v_k i_k = 0$, even when $\mathbf{v}$ and $\mathbf{i}$ come from two different circuits with the same graph.